Indian Institute of Engineering Science and Technology, Shibpur

B. Tech 8th Semester End-Semester Examination,

April 2025

Artificial Intelligence (CS4201)

Full Marks: 50 **Time: 3 Hours**

Answer Question No. 1 and any three Questions from the rest

1. (i) What do you mean by **zero sum game**?
   (ii) **"Confidence interval with a 95% confidence level"** – What does it mean?
   (iii) What is the difference between **immediate reward** and **cumulative reward**?

   2.5+2.5+3

2. (a) What are the *steps* involved in a typical *Reinforcement Learning* algorithm?
   (b) How to use the *Epsilon-Greedy* method to find the *Optimal Reward*?
   (c) Explain how Q network can get stuck in local optima, and what technique can be used to mitigate this issue.

   5+4+(2.5+2.5)

3. (a) Explain different phases of training to a Deep Q neural Network?
   Answer the question considering the training sample <S1, a2, R4, S7>.
   Assume Q values at different stages of execution.
   (b) Explain the role of the target network in DQN.

   10+4

4. (a) Formulate the following patterns as a constraint satisfaction problem.

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  S E N D
  M O R E
  =========
M O N E Y
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   (b) What do you mean by constraint propagation? What is soft constraint? 7+4+3

5. (a) Draw the semantic network that represents the data given below:
   • Mammals have fur. • All mammals are animals. • A bird is an animal. • A cat is a mammal.
   • Tom is a cat. • Tom is owned by John. • Tom is ginger in colour.
   (b) What are the different types of knowledge representation techniques used in solving AI problems? Give example for each of the cases.

   6+(4+4)

Indian Institute of Engineering Science and Technology, Shibpur

CST 8th Semester B Tech Examination, April 2025

Quantum Computing (CS-4226)

Full Marks 50 Time 3 hours

Answer Q.1 $((4+4+6)=14)$ and any four questions $((3+3+3)\times 4=36)$ from the rest.

1. (a) Describe an order finding algorithm and hence find the order of a mod 21, where you need to show all steps. **For odd roll nos $a = 11$ and for even roll nos $a = 19$.**
   (b) How can the order found above be used so that 21 can be factorized into its two prime factors?
   (c) Explain the role of quantum computing in such factorization.

2. (a) Describe an algorithm that utilizes quantum computing to find whether a Boolean function is balanced or constant.
   (b) Show how the time complexity improves upon employing quantum computing (over classical computing) in the above algorithm.
   (c) Describe how quantum search algorithm can be designed around this concept to achieve reduction of the search time.

3. (a) Describe the 3 qubit quantum Fourier transform (QFT) algorithm in terms of basic quantum gates and unitary transform.
   (b) Obtain the time complexity of QFT for problem size $N = 2^n$.
   (c) Show that the generalized QFT matrix is unitary.

4. (a) What is the notion of error that can occur in quantum channel?
   (b) Describe encoding scheme to tackle the errors in quantum channel using 3 qubits.
   (c) Show how the decoder can be designed to correct the designated errors.

5. (a) Describe any one of quantum computing implementation technologies.
   (b) Discuss about the merits and demerits of the chosen technology.
   (c) Describe some platform where quantum algorithms can be tested.

6. Discuss briefly about the following phenomena in the context of quantum computing:
   (a) Teleportation; (b) Quantum parallellism; (c) Swap gates.

7. Compare between the scenarios in classical and quantum computing:
   (a) Reversibility of gates;
   (b) Channel capacity bounds;
   (c) Observation and copying of information

INDIAN INSTITUTE OF ENGINEERING SCIENCE AND TECHNOLOGY SHIBPUR
SCHOOL OF MANAGEMENT SCIENCES

OPEN ELECTIVE, END SEMESTER (EVEN) EXAMINATION, APRIL 2025

SUBJECT NAME: ENTREPRENEURSHIP AND INNOVATION MANAGEMENT
SUBJECT CODE: MS4262

FULL MARKS: 50 Time: 3 hours

(Group- A & B should be answered on separate answer scripts)

GROUP –A

Marks: 30

Answer the Questions: (Any Three)

1. a) Differentiate Manufacturing and Service Enterprises.

(ii) Discuss the different Forms of Business Enterprises.

[3 + 7 = 10]

2. Present your hypothetical Business Plan, through a Business Model Canvas.

[10]

3. Discuss the Entrepreneurship Process; especially giving a focus on the process of assessing feasibility of the venture.

[10]

4. Case Study:

M. Thiagarajan, a pilot, was just 27 when he floated Paramount Airways, based out of Madurai, which launched commercial flights on 19th October 2005. This makes Thiagarajan the youngest airline CEO in the world. At the moment, Paramount Airways operates only in the south. Next year, Thiagarajan wants to launch it in the West. Two years after that, in 2008, he plans to enter the North and the East. 'We want to dominate the West by 2009 and become a national player by 2011', he says. A business management graduate, he entered the family business as a third-generation entrepreneur and established his own

Paramount Mills before starting the aviation business. If the low-profile entrepreneur has donned as aggressive profile, it is because 'we were making the right kind of noise and we are now leaders with 26 percent market share in the south Indian market', Thiagarajan says. 'Now it is time to have a presence in western and central India.' Thiagarajan may have eschewed a less-than flamboyant lifestyle in his personal life, but where Paramount Airways is concerned, he chose a 'high-value carrier business model targeting the premium segment of customers with a business ad first class configuration' using Brazil-made Embraer aircraft. More recently, Delhi- wk7 3 J1147 based MDLR Airlines has chosen to follow the model with an exclusive business class configuration.

(i) What are the typical traits of an entrepreneur that can be observed in M. Thiagarajan?

(ii) What do you see as the reason for the success of Paramount Airways?

[5+5]

GROUP –B

Marks: 20

Q1 is compulsory, and attempt any two from the rest (12+8=20 marks)

Q1. Case Study: EcoSoul – Powering Green Entrepreneurship through MSMEs

EcoSoul is a fast-growing micro, small, and medium enterprise (MSME) located in the heart of rural Uttar Pradesh. Founded in 2018 by two social entrepreneurs, the enterprise began with a mission to reduce India's plastic dependency by producing biodegradable and compostable tableware using agricultural by-products like sugarcane bagasse, rice husk, and wheat bran. What started as a modest operation with just 5 employees and one production unit has now grown into an award-winning sustainable brand that exports to over 15 countries, including the USA and Germany.

EcoSoul has created livelihood opportunities for over 300 rural women and marginal farmers by engaging them in the collection of agri-waste and initial stages of processing. It operates on a decentralized production model and reinvests a part of its profit into community education and green awareness campaigns. The

enterprise also adheres to fair-trade practices and actively measures its carbon footprint.

However, its journey hasn't been free of challenges. EcoSoul continues to grapple with poor access to formal credit, long processing delays for government subsidies under MSME schemes, lack of skilled manpower, and unreliable logistics support in rural areas. Despite being eligible for schemes like the Credit Guarantee Fund Trust for Micro and Small Enterprises (CGTMSE) and Zero Defect Zero Effect (ZED) certification, it often encounters bureaucratic hurdles and low policy awareness at the grassroots level.

Answer the following: (Total: 12 Marks):

a) Identify and explain the sustainability goals (from SDGs or ESG framework) that EcoSoul contributes to. (4)

b) Discuss the challenges faced by green MSMEs like EcoSoul in scaling up. (4)

c) Propose policy and institutional support that could enhance the growth and global reach of such enterprises under schemes like Startup India, Make in India, and Atmanirbhar Bharat. (4)

Attempt any two (Total: 4*2=8 Marks)

Q2. Discuss the triple-bottom-line approach in the context of Corporate Sustainability. Why is it important for CSR? (4)

Q3. Explain the relationship between Corporate Governance and CSR. How do they complement each other? (4)

Q4. Explain the concept of "Creative Destruction" as proposed by Joseph Schumpeter. How does this concept apply to the rise of innovative MSMEs in India, particularly in sectors like fintech, green tech, or digital services? (4)